

Intelligent Recruitment and Talent Analytics Platform

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Problem & Need

Existing Recruitment Challenges

- Large number of job applications.
- Manual resume screening is time-consuming.
- Difficulty in identifying suitable candidates quickly.
- Traditional recruitment processes may introduce bias.
- Limited use of analytics in recruitment decisions.

Need

An intelligent system that can:

- Automate resume screening.
- Match candidates with job requirements.
- Provide recruitment analytics.
- Support data-driven hiring decisions.

Objectives

- Automate resume processing and candidate screening.
- Match candidate skills with job requirements.
- Rank applicants based on suitability.
- Generate recruitment and talent analytics.
- Reduce recruiter workload and hiring time.
- Improve recruitment efficiency and decision-making.

Literature Review Overview

PAPER	AREA
Paper 1	AI & HR Analytics (2023)
Paper 2	Predictive HR Analysis (2024)
Paper 3	Resume NLP & Job Matching (2023)
Paper 4	Automated Resume Screening (2020)

Purpose of Review

- Understand existing recruitment technologies.
- Analyze strengths and limitations.
- Identify research gaps.
- Design an improved recruitment platform.

AI/HR Analysis

Focus

Application of AI and analytics in Human Resource Management.

Methods

- Data Analytics
- AI-based HR decision support
- Workforce data analysis

Key Findings

- Analytics improves recruitment decisions.
- HR can become more data-driven.
- Better workforce planning is possible.

Dataset

HR-related organizational data:

- Employee records
- Recruitment data
- Performance metrics
- Workforce information

Predictive HR Analytics

Focus

Using predictive analytics to improve workforce performance and HR decisions.

Methods

- Predictive Analytics
- Workforce Performance Analysis
- Decision Support Systems

Key Findings

- Predictive models improve HR planning.
- Better employee and workforce management.

Dataset

- Employee performance data
- Workforce management records
- HR operational data
- Historical employee information

Automated Resume Screening

Focus

Automated candidate screening using resume analysis.

Methods

- Resume upload.
- Resume parsing.
- Information extraction.
- Candidate filtering.
- Ranking and screening.

Dataset

- Collection of candidate resumes.
- Recruitment-related textual documents.
- Applicant information data.

Key Findings

- Faster shortlisting process.
- Efficient handling of large applicant pools.

Resume NLP & Job Matching

Focus

Resume Summarization and Job Description Matching.

Methods

- Resume collection.
- Resume text extraction.
- NLP preprocessing.
- Skill and information extraction.
- Job description analysis.
- Similarity score generation.
- Candidate-job matching.

Dataset

- Candidate resumes (PDF/Text)
- Job descriptions
- Skill-related textual data

Key Findings

- Automates resume analysis.
- Improves candidate-job matching
- Reduces recruiter effort.

Comparative Analysis

Feature	Paper 1 (P1)	Paper 2 (P2)	Paper 3 (P3)	Paper 4 (P4)
 HR Analytics	✓	✓ ✓	✗	✗
 Predictive Analytics	✓	✓ ✓	✗	Limited
 Resume Parsing	✗	✗	✓ ✓	✓
 Job Matching	Limited	✗	✓ ✓	✓
 Candidate Screening	Limited	✗	✓	✓ ✓
 Talent Analytics	✓	✓ ✓	✗	✗
 Complete Recruitment System	✗	✗	✗	✗



Strongly Addresses



Addresses

Limited

Partially Addresses



Not Addressed

Strengths & Limitations

Strengths

HR Analytics Papers

- Data-driven decision making.
- Workforce performance insights.
- Predictive capabilities.

Resume Screening Papers

- Faster screening.
- Reduced recruiter workload.
- Improved candidate matching.

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Limitations

- Lack of integrated recruitment workflow.
- Limited connection between screening and analytics.
- No unified recruitment management platform.

Research Gap

Existing Research

Group 1

HR Analytics → Decision Support

Group 2

Resume Screening → Candidate Matching

Gap Identified

No integrated platform combines:

- Resume Screening
- Candidate Matching
- Candidate Ranking
- Recruitment Management
- Talent Analytics
- Decision Support

How Our Project Addresses the Gap

Intelligent Recruitment and Talent Analytics Platform

Features:

- Resume Upload & Parsing
- NLP-Based Skill Extraction
- Candidate-Job Matching
- Candidate Ranking
- Recruitment Dashboard
- Talent Analytics
- Decision Support

Benefits: Combines screening and analytics into one platform.

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Thank You